What comes next?
Finding connections between word embeddings.

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1 Introduction

Embeddings of words are used, e.g. for text classification [Joulin et al. 2016] and for machine translations [Mikolov, Le, and Sutskever 2013]. Most important, they have proven to be able to capture semantic relationships such as analogies [Mikolov, Chen, et al. 2013]. Especially in the Digital Humanities, where word vector models are used to investigate the temporal dynamics of word meanings [Heuser 2016], word vectors enable a more complex semantic study as the computational semantics of the vector space relate to real-world semantics (or a fictonal world of literature). In the study of temporal changes in word meanings, usually a single vector model is trained for each period of time. Thus, the semantic shift is evaluated based on the shift of words compared to the next neighbouring vectors for each period of time.
Until today, the embeddings of different time steps are often trained independently and are only aligned with the help of the procrustes transformation afterwards [Kulkarni et al. 2015; Hamilton, Leskovec, and Jurafsky 2016; Szymanski 2017]. Only very recently, a method has been published that learns embeddings and alignment to its two temporal neighbours simultaneously and thus produces better results on temporal analogy tests [Yao et al. 2018].
However, to apply this method, knowledge about the true ordering is required. In cases where data does not follow a temporal dynamic, ground truth about this respective ordering might not be available.
In this paper, we present Word Embedding Networks (WEN), a new method that is substantially more flexible than those mentioned above and that allows the analysis of word dynamics in completely different regimes than the temporal one. So far, neighborhood between embeddings is explicitly given a priori, whereas in our method pairwise relations between embeddings and the embeddings themselves are learned simultaneously. This means that dynamics between any slicing of a text corpus can be learned (especially those where there is no order known) and the result not only contains embeddings for each slicing, but an order of slices that corresponds to the dynamics of meaning of the words.
The scope for the groupings is enormous which is especially interesting for the Digital Humanities with such a variety of meta data. For instance, the dynamics of meaning of words can not only be examined for different authors - it would also be conceivable to analyse the relationship of geographical groups of texts, different genres or writing devices, to name a few.

2 Methodology

RELATED WORK

The authors of [Yao et al. 2018] learned word embeddings for newspaper articles of the New York Times published between 1990 to 2016. Embeddings were trained individually for each year while ensuring that embeddings of neighboring years stay close together. The neighborhood constraint is manifested in the third summand of the objective function, i.e.

$$\frac{\tau}{2}\left(\|U_{t-1}-U_t\|_F^2+\|U_t-U_{t+1}\|_F^2\right) \tag{1}$$

thus knowledge about the true ordering is required. Here, $U_t \in \mathbb{R}^{V \times D}$ represents the D-dimensional word embeddings in a vocabulary of size V at time point t.

WORD EMBEDDING NETWORKS

For cases where the true ordering is unknown, we introduce a novel model that jointly orders the different slices while learning the individual embeddings. We therefore include an additional weight term $\omega_{t,t'}$ into our model. By updating $\omega_{t,t'}$ with respect to the distances between the corresponding word embeddings, it iteratively strengthens connections between embeddings that are close together, i.e. we optimize over the following function with gradient descent including momentum

$$\begin{aligned}\min_{U_t} F_t = \min_{U_t} & \frac{1}{2} \left\| Y_t - U_t U_t^\top \right\|_F^2 \\ & + \frac{\lambda}{2} \left\| U_t \right\|_F^2 \\ & + \frac{\tau}{2} \sum_{t' \neq t}^{N} \omega_{t,t'} \left(\left\| U_t - U_{t'} \right\|_F^2 \right).\end{aligned}$$

As before, U_t represents word embeddings at slice t and Y_t contains the PPMI matrix [Yao et al. 2018]. The first part of the optimization function handles training of the word embeddings while the second part ensures their sparse representation. The corresponding gradient w.r.t. U_t is composed as follows

$$\begin{aligned}\frac{\partial F_t}{\partial U_t} = & -(Y + U_t U_t^\top) U_t \\ & + \lambda U_t \\ & + \tau \sum_{t' \neq t}^{N} \omega_{t,t'} (U_t - U_{t'})\end{aligned}$$

and $\omega_{t,t'}$ will be updated following this update rule:

$$\begin{aligned}D &= \text{normalize} \left(\frac{1}{\text{dist } (U_t, U_t^\top)} \right) \\ \omega_{t,t'}^{\text{new}} &= \omega_{t,t'} + D \\ \omega_{t,t'}^{\text{new}} &= \text{normalize } (\omega_{t,t'}^{\text{new}})\end{aligned}$$

where

$$\text{normalize } (\omega_{t,t'}) = \frac{\omega_{t,t'}}{\left(\sum_i \omega_{ti} + \sum_i \omega_{it'} \right)}.$$

3 Results

3.1 Reconstructing known relationships

We first show how the relationships between the embeddings can be found if the underlying sequence of the slicing is already known. For this purpose, we use the New York Times data set mentioned before with filtered stop words. In this experiment, the information about the order of the years is withheld so that we can verify afterwards whether WEN could find relationships between neighbouring years.
To optimize the hyperparameter we divide the data set into two equal chunks and apply a grid search on the first half (1990 - 2003), which searches over the embedding dimensions and pairs of τ and λ. Our results show the same optimal parameters for τ and λ as reported by Yao et al. 2018 ($\lambda = 10$, $\tau = 50$) and an embedding dimension of 32. Applying those parameters to the second half of the data set we observe that 69% of the possible closest pairwise neighborhoods between the embeddings actually point to the embedding of the neighboring year.

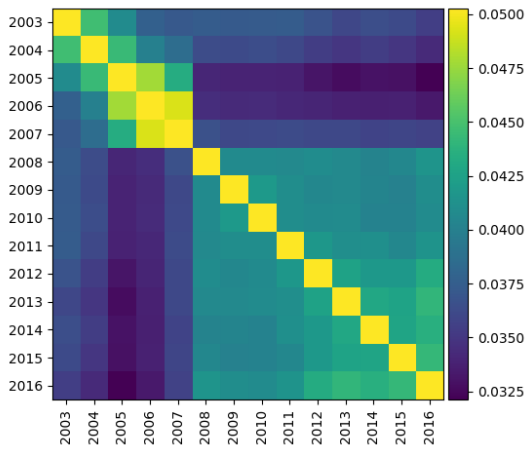
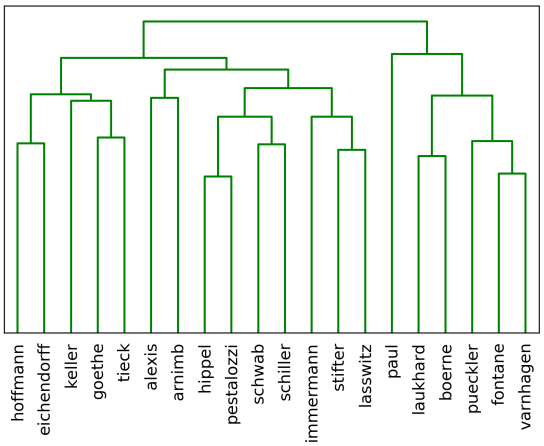


Figure 1: The off diagonal entries visualize ω after 3500 rounds of updates. Higher values relate to close distances between embeddings. It is clearly visible that the values get smaller whenever the distance between the years increase. The diagonal is set to the highest value (which would correctly being set to $+\infty$). Also, there seems to be another high level structure that groups the embeddings into two clusters (divided between 2007 and 2008).

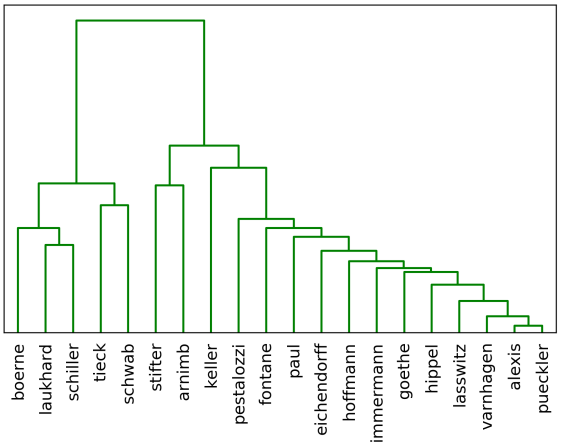
The learnt relationships between annual slices are visualized in Figure 1. From this relationship matrix between the embeddings of the years it is evident that a small temporal distance is accompanied by a high ω — with a few exceptions the correct order of the years can be read from the graph.

3.2 Relationships between authors in comparison with Burrow's Delta

As a second experiment we investigate the relationships between the embeddings of slices of texts in which the order does not exist as the underlying truth. For this purpose we use a part of the texts from the lemmatized versions of DTA (Deutsches Textarchiv), namely the 20 most represented authors of fiction texts (marked as "Belletristik") from 1770 to 1900 - in total this subcorpus comprises 10,219,336 lemmas.
In Figure 2a the results of the experiment are displayed in the form of a dendrogram. The results of Burrow's Delta on the same data [Burrows 2002] is shown in Figure 2b as a reference method. The linkage of the distance matrix for both dendrograms is performed with an averaging criterion. It is noticeable that Burrow's Delta learns several subtrees, which are then joined together at great merge costs. With WEN, however, the merge costs are much more uniformly distributed. This shows that WEN learns a smooth relationship between embeddings that evolve from each other. An example in which the two methods tend to agree is the early clustering of Immermann and Lasswitz, Laukhard and Boerne as well as Hoffmann and Eichendorff. However, the results differ for Tieck and Goethe, to give an example. The biggest difference between the two methods is that WEN also learns word representations that can be investigated further, where Burrow's Delta mostly uses the frequencies of stop words to differentiate authors that do not incorporate semantics.



(a) DYN shows smooth merge costs.



(b) Burrow's delta with 200 most frequent lemmas reveals sub-trees.

Figure 2: Comparison between WEN and Burrow's delta on DTA sliced into fiction texts of 20 Authors between 1770 and 1900

The embedding of texts by Theodor Fontane has the greatest distance from the embedding of Bettina von Arnim, hence there are presumably huge differences between the meaning of words. In Figure 3 a t-SNE embedding [Maaten and Hinton 2008] of the vectors of the word "Krieg" for Fontane (red) and Arnim (blue) are given and for both, the 19 most similar words of the corresponding embedding as an example explorative analysis. Apparently, the meaning of "Krieg" is very different for these two authors - when for Fontane it seems closely related to "Heer", "Schlacht", "Kampf" und "siegreich", for Arnim it reveals close words like "Angst", "fort", "vorbei".

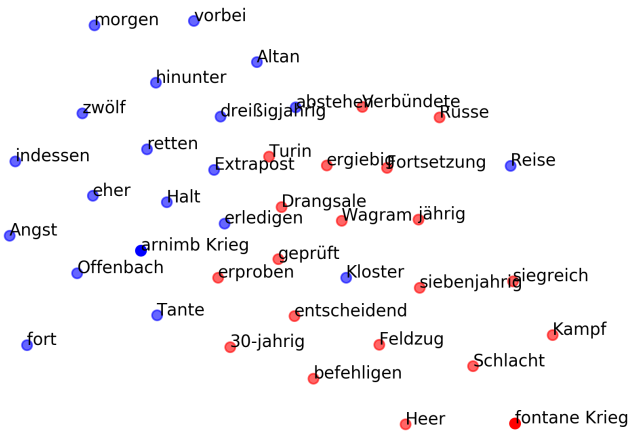


Figure 3: Two-dimensional t-SNE Embedding of the most similar vectors of "Krieg" in the Embedding of Bettina von Arnim and Theodor Fontane.

4 Conclusion

We introduced a novel method that learns orderings of arbitrary slices of text and showed that this method is particularly useful for the digital humanities. The method is not meant as a stylometric measure but rather as an explorative tool for semantic dynamics of words. This method will, by definition, likely find connections between any slices of a corpus. Even in the case of a heterogenous corpus for which an expert would deny any proximities between slices, WEN may converge to an optimum that yields a certain connection network. It is thus crucial to apply this method in close collaboration with domain experts and perform as many plausibility tests on individual words as possible. It may also be advising to probe stability across multiple random intitalizations of the weight matrices. When this advice is heeded WEN will serve as a new powerful tool for exploring the dynamics of word semantics.

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